

Question 1: Explain in your own words:

- What is Data Science?
- Why is it more than just statistics or programming?

Support your answer with one real world example.

Answer:

- 1 Data science is a multidisciplinary field that uses scientific methods, algorithms, and systems to extract meaningful insights and knowledge from structured and unstructured data.
- 2 Data Science is much more than just statistics (mathematics) or programming (coding) because it requires the ability to turn messy, real-world data into actionable business value.

Question 2: A retail company wants to:

- Collect customer data
- Build a churn prediction model
- Deploy it in production

Explain the role of:

- Data Analyst
- Data Scientist
- Data Engineer
- ML Engineer

At which stage does each contribute?

Answer:

- 1. Data Engineer**
 - **Stage:** Data Collection & Storage.
 - **Role:** They collect raw customer data from different sources and build data pipelines to store it in a database.
- 2. Data Analyst**
 - **Stage:** Data Cleaning & Insights.
 - **Role:** They clean the data and create reports or dashboards to understand customer behavior and current churn trends.
- 3. Data Scientist**
 - **Stage:** Model Development.
 - **Role:** They use Machine Learning algorithms to build a prediction model that can identify which customers are likely to leave.
- 4. ML Engineer**
 - **Stage:** Deployment & Production.

- **Role:** They put the model into the live production environment and ensure it works fast and reliably within the company's app or system.

Question 3: A hospital wants to:

- Predict patient readmission risk
- Automate report generation
- Deploy a chatbot for patient FAQs

Map each problem to:

- Traditional Data Science
- Machine Learning
- GenAI
- Agentic AI

Explain your reasoning.

Answer:

1. Predict patient readmission risk →

2. Machine Learning

- **Reasoning:** Predicting the future based on past data (like patient history) is a classic
- **Machine Learning** task. It uses patterns to calculate risk scores.

3. Automate report generation →

4. Traditional Data Science

- **Reasoning:** Creating reports from structured data involves data cleaning, calculation, and visualization. This falls under
- **Traditional Data Science** (using SQL, Excel, or Python).

3. Deploy a chatbot for patient FAQs →

GenAI

- **Reasoning:** A chatbot that understands and talks like a human to answer questions is a core application of
- **GenAI** (Generative AI), like ChatGPT.

5. Monitoring and Scheduling (Advance Step) →

6. Agentic AI

- **Reasoning:** If the system needs to take "actions" (like automatically booking an appointment or calling a doctor based on risk), it is called
 - **Agentic AI** because it acts as an independent agent.
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Question 4: An e-commerce company wants an AI system that:

- Monitors inventory
- Predicts demand
- Automatically places supplier orders
- Negotiates price via email

Explain:

1. Why this is an example of Agentic AI.
2. What risks might arise?
3. What human oversight would still be required?

Answer:

1. Why Agentic AI?

- **Reason:** Because the system
- **takes independent actions.** It doesn't just show data; it makes decisions and completes tasks like ordering and emailing by itself.

2. What are the Risks?

- **Wrong Orders:** The AI might order too much stock by mistake (financial loss).
- **Bad Negotiation:** It might agree to a very high price with the supplier.
- **System Errors:** Technical bugs could stop all orders, leading to zero stock.

3. Human Oversight Required:

- **Final Checks:** A human must approve very expensive or large orders.
- **Setting Rules:** Humans need to set limits (like a maximum budget) for the AI.
- **Monitoring:** Regular checking to ensure the AI is following company goals.

Question 5: Given:

price = 499

discount = 0.15

final_price = price - price * discount

2. Modify the code so that final_price is printed with 2 decimal places.

3. Convert price into a string and concatenate it into:

"The final price is ₹___"

```
price = 499
discount = 0.15
final_price = price - (price * discount)

# Using format to print 2 decimal places
print(f"{final_price:.2f}")

price_str = str(price)
message = "The final price is ₹" + price_str
print(message)
```

Question 6: A company gives bonus based on performance score:

- Score ≥ 90 : 20% bonus
- Score ≥ 75 : 10% bonus
- Else : No bonus

Write Python logic using:

- Comparison operators
- Logical operators
- Conditional statements

Answer:

```
score = 85 # Aap yahan koi bhi score likh sakte hain

if score >= 90:
    print("Bonus: 20%")
elif score >= 75:
    print("Bonus: 10%")
else:
    print("No bonus")
```

Question 7: You are given a customer review

```
review = "The product quality is Excellent but delivery was Late"
```

Write Python code to:

1. Convert to lowercase
2. Count number of words
3. Check if the word "excellent" exists
4. Replace "Late" with "delayed"

Answers:

```
review = "The product quality is Excellent but delivery was Late"

# 1. Convert to lowercase
lower_review = review.lower()
print(lower_review)

# 2. Count number of words
word_count = len(review.split())
print(word_count)

# 3. Check if the word "excellent" exists
exists = "excellent" in lower_review
print(exists)

# 4. Replace "Late" with "delayed"
new_review = review.replace("Late", "delayed")
print(new_review)
```

Question 8: Given sales data:

```
sales = [12000, 15000, 11000, 18000, 17000]
```

1. Calculate total revenue.
2. Find average revenue.
3. Identify highest sales month.
4. Append new month sale = 20000.
5. Sort the list in descending order

Answers:

```
sales = [12000, 15000, 11000, 18000, 17000]

# 1. Calculate total revenue
total = sum(sales)
print("Total Revenue:", total)

# 2. Find average revenue
average = sum(sales) / len(sales)
print("Average Revenue:", average)

# 3. Identify highest sales
highest = max(sales)
print("Highest Sale:", highest)

# 4. Append new month sale (20000)
sales.append(20000)
print("Updated List:", sales)

# 5. Sort the list in descending order (Bade se chota)
sales.sort(reverse=True)
print("Sorted Sales:", sales)
```

Question 9: You are building a simple sales analysis system.

Given:

```
products = ["Laptop", "Phone", "Tablet"]
prices = [50000, 20000, 15000]
quantities = [5, 10, 7]
```

Tasks:

1. Compute total revenue for each product.
2. Identify which product generated maximum revenue.
3. Store revenues in a new list.
4. Print a formatted summary like:

```
products = ["Laptop", "Phone", "Tablet"]
prices = [50000, 20000, 15000]
quantities = [5, 10, 7]

# 1 & 3. Compute total revenue and store in a new list
revenues = []
for i in range(len(products)):
```

```
rev = prices[i] * quantities[i]
revenues.append(rev)

# 2. Identify product with maximum revenue
max_rev = max(revenues)
max_index = revenues.index(max_rev)
best_product = products[max_index]

# 4. Print a formatted summary
print("--- Sales Summary ---")
for i in range(len(products)):
    print(f"{products[i]}: ₹{revenues[i]}")

print(f"\nProduct with Max Revenue: {best_product} (₹{max_rev})")
```

Question 10: A startup collects customer feedback as text and numerical ratings.

Explain step-by-step:

1. How raw text becomes structured data.
2. Where Python (strings & lists) is used.
3. Where Data Science models are used.
4. Where GenAI can enhance the system.
5. What ethical concerns must be addressed.

Answers:

1. Raw text to Structured data

- **Step:** Raw feedback (sentences) is cleaned by removing extra spaces and special characters. It is then converted into a table format (like Excel) with columns for "Date," "Rating," and "Keywords."

2. Use of Python (Strings & Lists)

- **Strings:** Used to clean and handle the text feedback (e.g., converting to lowercase).
- **Lists:** Used to store multiple ratings or a collection of customer comments for processing.

3. Use of Data Science Models

- **Purpose:** To perform **Sentiment Analysis** (identifying if feedback is Happy, Sad, or Angry) and to predict future ratings based on current trends.

4. How GenAI enhances the system

- **Action:** GenAI can automatically **summarize** thousands of long reviews into a short paragraph and write polite, automated replies to customers.

5. Ethical Concerns

- **Privacy:** Customer names and personal details must be kept secret (Data Privacy).
- **Bias:** The AI should not be unfair to any specific group of people based on their language or background.